MATH70110: QRM Coursework 1

Anandakrishnan Suresh[*], Eugene Lee[†], Rishi Massey[‡], Alexandros Violetti[§]

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1 Abstract

We follow the general instruction of the course to share the results and make extensions to complement our findings. In part A, we observe stylized facts of financial return in EURO STOXX 50, and fit conditional variance models accordingly. We iteratively improve the model starting from GARCH(1,1)-normal, and conclude that ARMA terms are necessary by comparing ARMA-GARCH-t and GARCH-t models with the likelihood ratio test. In part B, we discover that both FHS-EWMA and FHS-GARCH forecasting pass the standard regulatory backtesting practices. We go on to perform Diebold-Mariano test to demonstrate FHS-GARCH has better predictive power over a consistent scoring, and recommend Bob to use FHS-GARCH if he prioritizes forecast accuracy, and FHS-EWMA if he prefers its simplicity.

2 Part A: EURO STOXX 50

2.1 Empirical Analysis of Log Returns (i and ii)

We report our findings on empirical log returns over EURO STOXX 50 along with plots and tables to support them. The returns are all in-line with stylized facts of financial return covered in our module. Exact values can be found in table 1.

- The distribution of log-returns is leptokurtic, and its kurtosis exceeds 13.
- The mean of the distribution is near-zero.
- The distribution is asymmetric, specifically negatively skewed.
- The distribution is non-normal.

[*]CID 06042471
[†]CID 06049056
[‡]CID 02218595
[§]CID 06048602

Moment	Estimate
Mean (%)	0.0118
Std (%)	1.2391
Skewness	-0.7960
Kurtosis	13.2781

Table 1: Summary moments of SX5E daily log-returns (in percent). [**Appendix 1**]

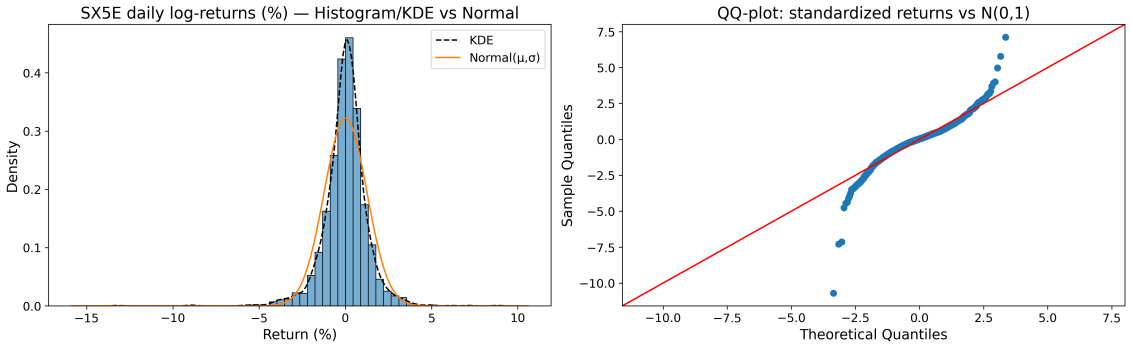


Figure 1: Plot to observe goodness of fit. Histogram on the left verifies the moments we computed: actual distribution is more peaked in the centre compared to normal, tail-heaviness (kurtosis greater than 10) as the histogram crosses normal distribution with outliers, left-skewed (skew less than 0) with more coverage towards the left of center. QQ-plot on the right exhibits deviation away from normal towards tail-end. [**Appendix 1**]

The ACF of raw returns shows no significant autocorrelation, which indicates that returns behave like white noise. In contrast, the ACFs of absolute and squared returns display clear and persistent autocorrelation across many lags. This pattern reflects volatility clustering: periods of high volatility tend to follow high-volatility periods, and quiet periods tend to follow quiet periods. These features imply time-varying variance, so a conditional variance model such as GARCH is appropriate. Figure 2 shows that returns over 50-lag are uncorrelated, whereas absolute returns and squared returns signal clustering by crossing the significance threshold.

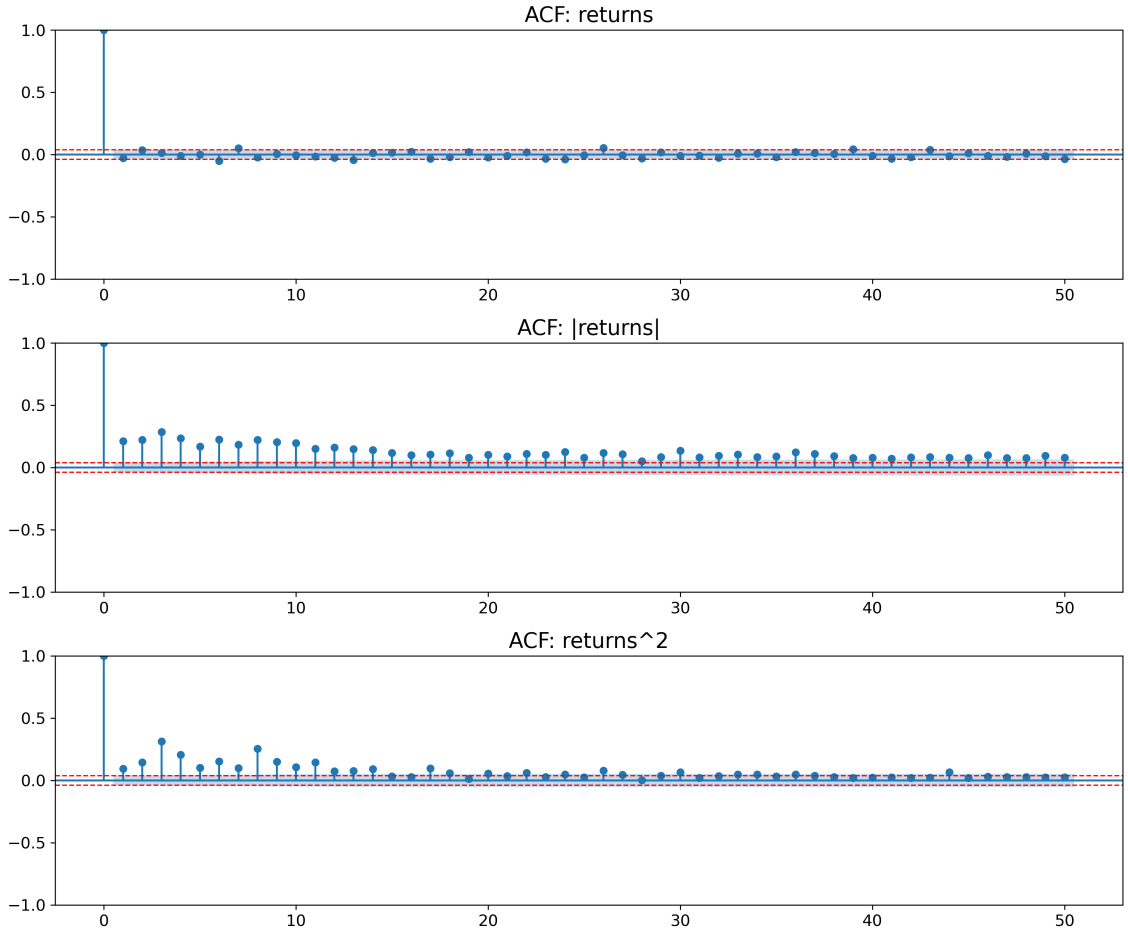


Figure 2: ACF plotted for returns, absolute returns, and squared returns. The red dotted line indicate band of significance given by 1.96/sqrt(N), N is the number of observations (2505). **[Appendix 2]**

2.2 Volatility Modelling and Model Evaluation (iii, iv, v)

To account for conditional variance, we fit a GARCH(1,1) normal model to the log returns to obtain the following summary along with diagnostics. Table 2 summarizes the estimates of the coefficients, figure 3 plots the scaled absolute returns along with fitted volatilities. We used arch python library to fit the model. Here's the notation we use throughout for parameters of GARCH(1,1)-normal model.

The mean equation is given by:

$$r_t = \mu + \varepsilon_t,$$

where the innovations follow:

$$\varepsilon_t = \sigma_t z_t, \qquad z_t \sim \mathcal{N}(0,1).$$

The conditional variance follows a GARCH(1,1) process:

$$\sigma_t^2 = \omega + \alpha_1 \varepsilon_{t-1}^2 + \beta_1 \sigma_{t-1}^2.$$

Parameter	Estimate
μ	0.0476
ω	0.0596
α_1	0.1390
β_1	0.8243

Table 2: GARCH(1,1)-Normal Model Fit Parameters. **[Appendix 3]**

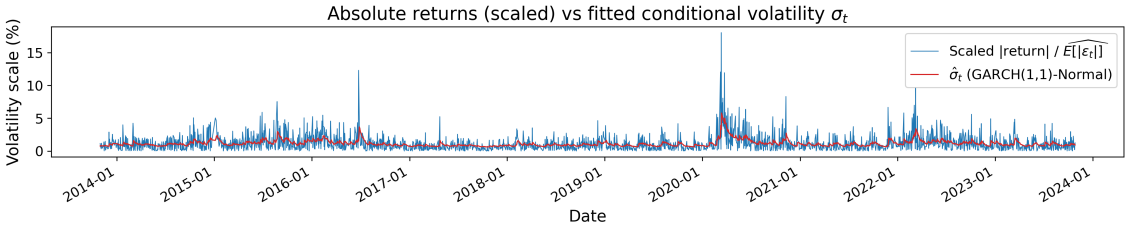


Figure 3: Absolute returns scaled with respect to mean innovation plotted together with volatility fit by GARCH(1,1)-Normal model. We can visually see the volatility terms following the general trend, tracking volatility clusters and persistence. **[Appendix 3]**

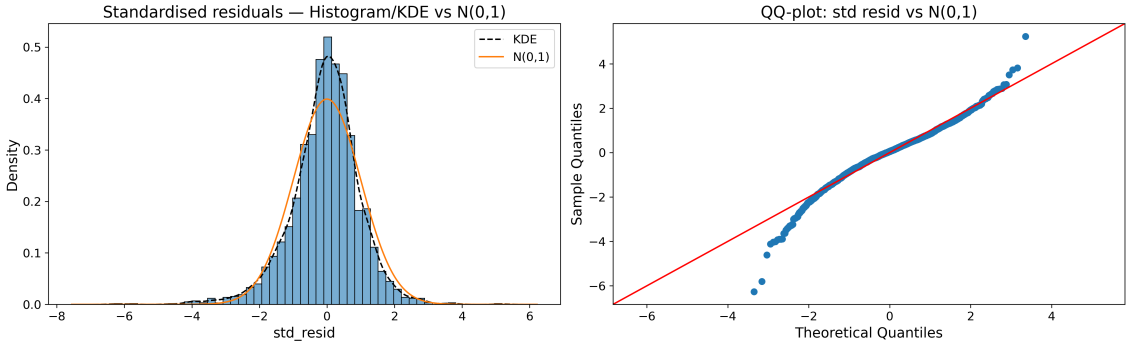


Figure 4: Standardized residual analysis. The histogram and KDE show that the standardized residuals are more peaked and have heavier tails than a normal N(0,1) distribution. The QQ-plot confirms this: the middle quantiles track the 45-degree line reasonably well, but the extreme quantiles deviate upward and downward, indicating tail-heaviness. **[Appendix 4]**

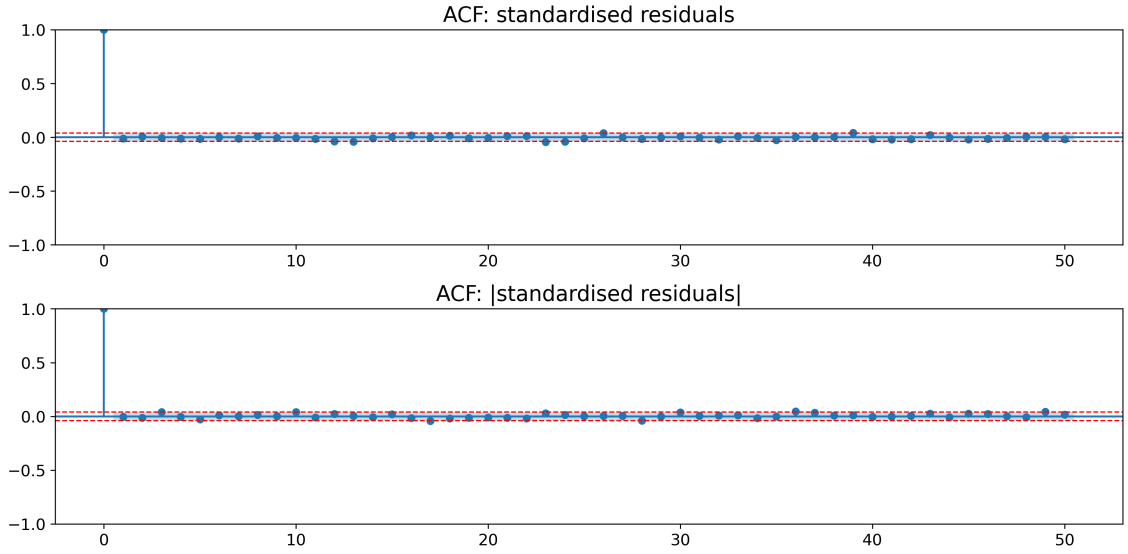


Figure 5: ACF of standardized residuals and absolute standardized residuals. We notice that both the raw and absolute values are uncorrelated. [**Appendix 4**]

At an outset, the fitting looks reasonable in figure 3, and the combined coefficient exceeds 0.9, which could be interpreted as high persistence of volatility. But we notice from figure 4 that the residuals are not normal. Although figure 5 shows no visible autocorrelation, the ACF alone cannot establish full independence.

Because the standardized residuals from the GARCH(1,1)-Normal model remain heavy-tailed and show deviations from normality, the model underestimates extreme behavior. To account for this model misspecification, we fit ARMA-GARCH-t model as an alternative, with the hopes of student-t and ARMA mitigating this issue. Since the arch library does not support ARMA GARCH-t out of the box, we adapted the implementation from the module's Part 3 lecture notes (Gonon & Coache 2025). [**Appendix 5**]

To revisit the definitions and notations, the conditional mean follows an ARMA(1,1) structure:

$$r_t = \mu + \phi_1 r_{t-1} + \theta_1 \varepsilon_{t-1} + \varepsilon_t.$$

The innovations are:

$$\varepsilon_t = \sigma_t z_t, \qquad z_t \sim t_\nu(0,1),$$

where ν denotes the degrees of freedom.

The conditional variance follows a GARCH(1,1) process:

$$\sigma_t^2 = \alpha_0 + \alpha_1 \varepsilon_{t-1}^2 + \beta_1 \sigma_{t-1}^2.$$

This leads to a couple of questions 1) Is ARMA-GARCH-t better than GARCH-normal? 2) Is the ARMA term necessary? We note the near-cancellation of ϕ_1 and θ_1 terms on table 3 suspect ARMA's efficacy. The answer to the first question is fairly simple compared to the latter. Table 4 displays likelihood and both information criteria favoring ARMA-GARCH-t, with AIC and BIC improving from 7485.84 and 7509.14 to 7284.18 and 7324.96 respectively.

Quantity	Value
α_0	0.0475
α_1	0.1477
β_1	0.8336
μ	0.0699
ϕ_1	0.9115
θ_1	-0.9336
ν (df)	4.7273

Table 3: ARMA-GARCH-t Model Fit Summary **[Appendix 5]**

The second question, however, isn't as easy because we cannot explain whether the apparent performance gain is attributed to the ARMA term or the student-t. To answer this question, we fit GARCH-t to compare against ARMA-GARCH-t and perform a likelihood ratio test on nested models (Wasserman 2004, p. 164).

Hypothesis. The null hypothesis is

$$H_0 : \ \phi_1 = 0, \quad \theta_1 = 0,$$

meaning that including ARMA structure in the conditional mean does *not* improve model fit. The alternative hypothesis is

$$H_1 : \ (\phi_1, \theta_1) \neq (0, 0),$$

meaning that the AR and MA terms *do* improve predictive performance.

Test Statistic. The likelihood ratio test statistic is computed as

$$LR = 2\left(\ell_{\text{full}} - \ell_{\text{restr}}\right),$$

where ℓ_{full} is the log-likelihood of the ARMA-GARCH–t and ℓ_{restr} is the log-likelihood of the constant-mean GARCH-t. Since the unrestricted model nests the restricted model, the statistic satisfies

$$LR \sim \chi^2_q,$$

where the number of restrictions is

$$q = 7 - 5 = 2.$$

Decision Rule. Reject the null hypothesis if

$$p = 1 - F_{\chi^2_2}(LR) < 0.05.$$

Interpretation:

- $p < 0.05$: the ARMA terms significantly improve model fit.
- $p \geq 0.05$: the ARMA terms are unnecessary.

We report the results in table 4 and 5, and conclude that **ARMA terms are necessary**. Student-t seems to be the main factor of improvement, as we observe significant information criteria improvement between GARCH-normal and GARCH-t. On one hand, we do have doubts in the value of the ARMA term in context of financial modelling due to the questionable magnitude of AIC/BIC improvements (and the fact that it wasn't available in a popular library, hinting it's not a common practice). But we decide that the added complexity of 2 terms are statistically significant by testing.

Model	Log-Likelihood	AIC	BIC
GARCH(1,1)–Normal	-3738.92	7485.84	7509.14
ARMA–GARCH–t	-3635.09	7284.18	7324.96

Table 4: Model Comparison: GARCH(1,1)-Normal vs. ARMA-GARCH-t [**Appendix 3**] [**Appendix 6**]

Model	Log-Likelihood	AIC	BIC
ARMA–GARCH–t (Full)	-3635.092	7284.184	7324.964
GARCH–t (Restricted)	-3639.079	7288.159	7317.287
Likelihood Ratio Test: $H_0 : \phi_1 = \theta_1 = 0$			
LR = 7.974 df = 2 p-value = 0.0186			

Table 5: Likelihood Ratio Test and Information Criteria Comparison [**Appendix 7**]

3 Part B: Bob's Big Short

3.1 VaR and ES forecast Results

This section presents the one-day-ahead Value-at-Risk (VaR) and Expected Shortfall (ES) forecasts generated using three methods: (i) Historical Simulation (HS), (ii) Filtered Historical Simulation (FHS) with EWMA volatility, and (iii) FHS with a GARCH(1,1) volatility model. For each method, both the 95% and 99% VaR and ES forecasts are computed using a rolling window of the previous 500 daily observations. Figure 6 displays the realised positive losses together with the forecasted VaR and ES series for each method.

3.1.1 (i) Historical Simulation (HS)

Historical Simulation constructs VaR and ES directly from the empirical distribution of the past 500 daily losses. For each day t, the most recent 500 observations are sorted, and the empirical α-quantile is taken as the VaR, while ES is obtained as the average of the worst α fraction of outcomes. Since HS uses raw historical returns without modelling time-varying volatility, the resulting VaR and ES series exhibit step-like behaviour and respond slowly to changes in market conditions. Forecasts remain constant for extended periods, moving only when large returns enter or exit the rolling window. As a result, HS tends to underestimate risk during periods of rapidly rising volatility and overestimate for a certain time after a significant loss event.

Note: Determining the Distribution F_Z in Filtered Historical Simulation

In both FHS–EWMA and FHS–GARCH, the return process is decomposed as

$$r_t = \sigma_t Z_t,$$

where σ_t is the conditional volatility estimate and Z_t is the standardised innovation. After computing σ_t using either EWMA or GARCH, the shocks are obtained as

$$Z_t = \frac{r_t}{\sigma_t}.$$

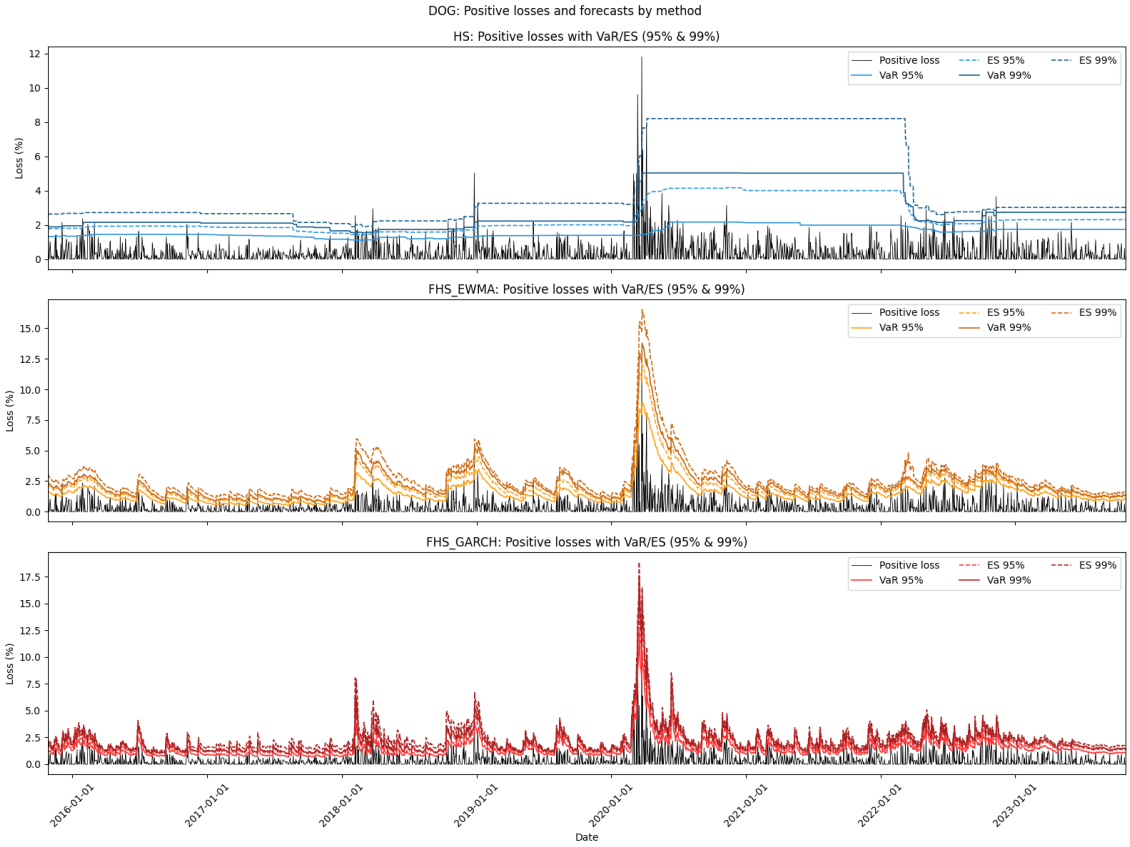


Figure 6: Positive losses with VaR and ES forecasts using HS, FHS-EWMA, and FHS-GARCH. **[Appendix 8]** **[Appendix 9]**

To determine the distribution F_Z, a rolling window of the most recent 500 standardised residuals is collected:

$$Z_{t-500+1}, \ldots, Z_t.$$

The empirical distribution of these residuals is used as an estimate of F_Z:

$$F_Z(z) = \frac{1}{500} \sum_{i=1}^{500} \mathbf{1}\{Z_i \leq z\}.$$

The one-day-ahead VaR and ES forecasts are then obtained by taking the empirical quantiles of this distribution and rescaling them using the forecasted conditional volatility:

$$VaR_{\alpha,t+1} = \sigma_{t+1} \cdot q_\alpha(Z), \qquad ES_{\alpha,t+1} = \sigma_{t+1} \cdot E[Z \mid Z \leq q_\alpha(Z)].$$

Thus, determining F_Z amounts to estimating the empirical distribution of volatility-adjusted innovations, from which the tail risk measures are derived.

3.1.2 (ii) Filtered Historical Simulation with EWMA (FHS-EWMA)

To incorporate time-varying volatility, the EWMA approach estimates conditional variance using the recursion

$$\hat{\sigma}^2_{t+1} = \alpha(X_t - \hat{\mu}_t)^2 + (1-\alpha)\sigma^2_t$$

with smoothing parameter $\alpha = 0.06$ as given (by convention). Returns are standardised using σ_t, and Historical Simulation is applied to these filtered residuals over the latest 500 observations to obtain an empirical distribution of shocks. The one-day-ahead VaR and ES forecasts are then rescaled using the EWMA volatility forecast. Compared with HS, the FHS-EWMA method produces smoother and more responsive VaR and ES series, with forecasted risk levels increasing and decreasing more gradually in line with recent volatility dynamics.

3.1.3 (iii) Filtered Historical Simulation with GARCH(1,1) (FHS-GARCH)

The FHS-GARCH approach replaces the EWMA volatility filter with a GARCH(1,1) model, estimated each day using the most recent 500 returns. The recursion follows as

$$\hat{\sigma}_{t+1}^2 = \alpha_0 + \alpha_1(X_t - \hat{\mu}_t)^2 + \beta_1\hat{\sigma}_t^2,$$

allowing for volatility persistence and clustering. After standardising returns by the GARCH volatility estimates, Historical Simulation is applied to the rolling-window residuals, and the resulting empirical shocks are rescaled using the one-step-ahead GARCH volatility forecast. The resulting VaR and ES series respond sharply to volatility bursts and decay gradually during calmer periods, reflecting the mean-reverting nature of GARCH dynamics.

3.1.4 Comparison of Forecast Behaviour

The three forecasting methods display distinct behaviours. By referring back to Figure 6, we see that HS forecasts are relatively flat and slow to adjust, often failing to capture sudden increases in market volatility. FHS-EWMA introduces smoother and more timely adjustments in risk estimates by incorporating exponentially-weighted volatility updates. FHS-GARCH exhibits the strongest responsiveness to volatility clustering and produces risk forecasts that track market turbulence most closely. Overall, introducing a volatility filter greatly improves the timeliness and economic plausibility of the VaR and ES forecasts compared to basic Historical Simulation. We now progress to backtesting, where we run coverage and independence tests on each forecasting method, in order to draw statistical conclusions on accuracy.

3.2 Backtesting Results

In this section, we will cover the results of the backtesting for the three forecast methods (Historical simulation, Filtered historical simulation (FHS) with exponentially-weighted moving average (EWMA) and FHS with GARCH (1,1)). Since the metrics of interest are the risk measures, VaR and ES, the tests we conduct will analyze the legitimacy of our estimates.

For VaR, we will perform two tests. One, called the Kupiec's unconditional coverage test, assesses whether the observed frequency of VaR violations is in line with the confidence level (α) and two, called Joint test for coverage and independence, combines the Kupiec's coverage test with Christoffersen's independence test which detects whether the VaR violations occur in clusters ([**Appendix 10**]). The null hypothesis for the Kupiec's unconditional coverage test is stated as follows:

$$H_0 : E[I_t \mid \mathcal{F}_{t-1}] = 1 - \alpha, \quad t = 1, \ldots, T$$

where $I_t := \mathbf{1}_{\{L_t > \widehat{\mathrm{VaR}}_\alpha(L_t)\}}$, $\widehat{\mathrm{VaR}}_\alpha(L_T)$ is the forecast for the corresponding loss L_t, and $\mathcal{F}_{t-1}$ is the σ-algebra generated by $I_1, \ldots, I_{t-1}$. Similarly, the null hypothesis for Christoffersen's independence test is stated as follows:

$$H_0 : P[I_t = 1 \mid I_{t-1} = 1] = P[I_t = 1 \mid I_{t-1} = 0], \quad t = 2, \ldots, T$$

Thus, the joint test for coverage and independence can defined by simply combining the test statistics $\mathrm{LR}_{\mathrm{uc}}$ and $\mathrm{LR}_{\mathrm{ind}}$, for the unconditional coverage and the independence respectively, in the following way:

$$\mathrm{LR}_{\mathrm{joint}} = \mathrm{LR}_{\mathrm{uc}} + \mathrm{LR}_{\mathrm{ind}}$$

under the joint null hypothesis,

$$H_0 : P[I_t = 1 \mid I_{t-1} = 1] = P[I_t = 1 \mid I_{t-1} = 0] = 1 - \alpha, \quad t = 2, \ldots, T$$

Table 6 presents the results of the tests performed across the confidence levels (α) 95% and 99%. The violation rate is the proportion of times the actual loss exceeded the predicted VaR. Ideally, this should be close to $1 - \alpha$ meaning 5% and 1% for the 95% and 99% levels, respectively. From the table, all methods exhibit violation rates that are reasonably close to their corresponding confidence levels, which suggests that the VaR estimates are well-calibrated in general. The next columns, $\mathrm{LR}_{\mathrm{uc}}$ and p_{uc} represent the Kupiec's unconditional coverage test statistic and its p-value. At both confidence levels, the p-values are mostly above the $(1-\alpha)\%$ threshold, implying adequate coverage for most models, except the HS method at the 99% level. The subsequent columns, $\mathrm{LR}_{\mathrm{ind}}$ and $\mathrm{p}_{\mathrm{ind}}$, represent the test statistic and its p-value for Christoffersen's independence test. Similar to the coverage test, most of the models here showed significant evidence to not reject the null hypothesis, except the HS method at 95%, which showed some evidence of underestimation of risk during crises. The final columns, $\mathrm{LR}_{\mathrm{joint}}$ and $\mathrm{p}_{\mathrm{joint}}$, combine both tests to assess overall model adequacy. Consistent with the earlier observations, the FHS–EWMA model shows the most robust performance, maintaining high p-values across both confidence levels, indicating that it provides reliable and independent VaR forecasts.

Method	α	Violation rate	$\mathrm{LR}_{\mathrm{uc}}$	p_{uc}	$\mathrm{LR}_{\mathrm{ind}}$	p_{ind}	$\mathrm{LR}_{\mathrm{joint}}$	p_{joint}
HS	0.95	0.049	0.032	0.858	4.722	0.030	4.754	0.093
FHS EWMA	0.95	0.051	0.053	0.819	0.367	0.545	0.419	0.811
FHS GARCH	0.95	0.050	0.001	0.980	2.612	0.106	2.613	0.271
HS	0.99	0.015	4.228	0.040	3.056	0.080	7.284	0.026
FHS EWMA	0.99	0.011	0.389	0.533	0.531	0.466	0.921	0.631
FHS GARCH	0.99	0.014	3.457	0.063	0.847	0.357	4.304	0.116

Table 6: Backtesting Results for VaR Forecast Methods. [**Appendix 11**]

To test ES, the most common method is to measure whether the losses are systematically greater than the ES forecasts on days when VaR violation occurs ([**Appendix 10**]), which leads to the null hypothesis:

$$H_0 : \widehat{\mathrm{ES}}_\alpha(L_t) = \mathrm{ES}_\alpha(L_t), \quad \widehat{\mathrm{VaR}}_\alpha(L_t) = \mathrm{VaR}_\alpha(L_t), \quad t = 1, \ldots, T$$

where $\widehat{\mathrm{ES}}_\alpha$ represents the expected shortfall estimated under the corresponding $\widehat{\mathrm{VaR}}_\alpha$ forecast.

Table 7 reports the backtesting results for the ES forecasts under both the 95% and 99% confidence levels. At the 95% confidence level, all three methods produce p-values comfortably above the 5% significance threshold, suggesting reliable ES estimates by the models. At the 99% confidence level, the results remain broadly consistent. Both FHS approaches continue to produce significant evidence not to reject the null hypothesis. The only exception is the HS method, which yields a p-value of 0.038, just below the typical 5% threshold, suggesting some underestimation of risk in the far tail. Overall, the filtered historical simulation methods deliver more accurate and robust ES forecasts across both confidence levels.

Method	α	p-value
HS	0.95	0.130
FHS EWMA	0.95	0.717
FHS GARCH	0.95	0.209
HS	0.99	0.038
FHS EWMA	0.99	0.766
FHS GARCH	0.99	0.621

Table 7: Backtesting Results for ES Forecast Methods. [**Appendix 11**]

3.3 Diebold-Mariano Test

Since both FHS models pass the standard regulatory backtests, we introduce statistical tools to convince Bob that one model is better than the other. The Diebold–Mariano test compares the accuracy of the forecast of two competing forecasts over a scoring function (Diebold & Mariano 1995).

Let $L_{\text{EWMA},\,t}$ and $L_{\text{GARCH},\,t}$ be the loss of FHS-EWMA and FHS-GARCH respectively.

$$d_t = L_{\text{EWMA},\,t} - L_{\text{GARCH},t}, \quad t = 1, \ldots, T,$$

Hypothesis. The null hypothesis is

$$H_0 : \; E[d_t] = 0,$$

meaning that both models have equal predictive accuracy. The expected loss differential between their forecasts is zero. The alternative hypothesis is

$$H_1 : \; E[d_t] \neq 0,$$

implying that one of the models yields systematically lower forecast loss and thus superior predictive performance.

Test Statistic. Test statistic is computed as

$$DM = \frac{\bar{d}}{\sqrt{\frac{\hat{\gamma}_d(0) + 2\sum_{k=1}^{h-1}\hat{\gamma}_d(k)}{T}}},$$

where

$$\bar{d} = \frac{1}{T}\sum_{t=1}^{T} d_t$$

is the sample mean of the loss differentials, $\hat{\gamma}_d(k)$ denotes the sample autocovariance of d_t at lag k, and h is the forecast horizon (with $h = 1$ for daily one-step-ahead forecasts). We note that our forecast time-horizon is only 1. Under H_0, the statistic approximately follows the standard normal distribution:

$$DM \sim N(0, 1).$$

Decision Rule. Reject the null hypothesis if

$$|DM| > z_{1-\alpha/2}$$

or equivalently if the p-value satisfies $p < \alpha$.

Interpretation:

- $p < 0.05$: the models differ significantly in forecast accuracy. The model with the lower mean loss provides superior forecasts.
- $p \geq 0.05$: no significant difference. Both models perform similarly.

To finalize the argument, we formalize the choice of loss function (or in this case, a score on functionals of distributions) specified in this test. (VaR_α, ES_α) pair are jointly elicitable, which means that there is a strictly consistent scoring function for (VaR_α, ES_α) forecast problems (Fissler & Ziegel 2016). We compute the trivial form of the general loss introduced in Corollary 5.5 of the mentioned paper.

$$L_t = (I_t - \alpha)\left(VaR_{\alpha,t} - Y_t\right) + \frac{I_t}{\alpha}\left(ES_{\alpha,t} - VaR_{\alpha,t}\right),$$

where

$$I_t = 1\{Y_t \leq VaR_{\alpha,t}\}$$

is an indicator that equals one if the realized loss exceeds the VaR estimate (i.e., a violation), and zero otherwise.

According to the results in Table 8, the FHS-GARCH model clearly outperforms FHS-EWMA, with a p-value effectively equal to zero due to the test statistic exceeding 12 standard deviations. The positive DM statistic indicates that FHS-GARCH produces lower average loss, and therefore more accurate forecasts. While this may seem like a strong result, it is not entirely unexpected since FHS-GARCH is a fully specified statistical model, whereas FHS-EWMA is more of a heuristic approach. In that sense, the comparison is not entirely even. Therefore, we conclude by recommending Bob to use FHS-GARCH if he values predictive accuracy and methodological rigour, and FHS-EWMA if he prefers a simpler, more transparent method.

Model Comparison	DM Statistic	p-value
EWMA vs. GARCH ($\alpha = 0.95$)	12.6800	negligible

Table 8: Diebold-Mariano test results comparing EWMA and GARCH using the FZ loss function. **[Appendix 12]**

Acknowledgments

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